	Parul University **Faculty of Engineering and Technology** **Parul Institute of Engineering and Technology** **Computer Science & Engineering(AI&DS) Department**

Name of the teacher: Deepak Palavalasa **Subject Name: 303105253 Software Engineering – 4AIDS4**		**Hours/Week: 3 Hours** **Mon, Tue, Wed**
Sr. No.	**Name of Topic**	**Planned Date**
CH 1	**Transformer Testing**	**06**
1	Study of Different Models, Software Characteristics, Components, Applications	27-11-2025
2	Layered Technologies, Processes, Process Models- Waterfall model, Incremental, Evolutionary process models- Prototype	27-11-2025
3	Spiral And Concurrent Development Model	28-11-2025
4	Agile Development :	28-11-2025
5	Agility and Agile Process model, Extreme	29-11-2025
6	Other process models of Agile Development and Tools.	04-12-2025
2	**Software Project Management:**	05
7	Management Spectrum, People 'Product ' Process- Project,	04-12-2025
8	W5HH Principle, Importance of Team Management	05-12-2025
9	Planning a Software Project : Scope and Feasibility, Effort Estimation,	11-12-2025
10	Schedule and staffing, Quality Planning, Risk management- identification,	12-12-2025
11	project monitoring plan, Detailed Scheduling	13-12-2025
3	**Requirements Engineering:**	05
12	Problem Recognition, Requirement Engineering tasks	18-12-2025
13	Processes, Requirements Specification,	19-12-2025
14	Use cases	26-12-2025
15	Functional specification,	27-12-2025
16	Requirements validation, Requirements Analysis	01-01-2026
4	**Structured System Design:**	05
17	Design Concepts, Design Model, Software Architecture, Data Design,	02-01-2026

18	Architectural Styles and Patterns, Architectural Design, Alternative architectural designs,	08-01-2026
19	Modeling Component level design and its modeling, Procedural Design, Object Oriented Design	09-01-2026
20	Data Oriented Analysis & Design : Difference between Data and Information, E-R Diagram,	10-01-2026
21	Dataflow Model, Control Flow Model, Control and Process Specification, Data Dictionary	16-01-2026
5	**Coding and Unit Testing:**	04
22	Programming principles and guidelines, Programming practices	22-01-2026
23	Management of code evaluation, Unit testing- procedural units,	23-01-2026
24	classes, Code Inspection, Metrics- size measure,	24-01-2026
25	complexity metrics, Cyclomatic Complexity, Halstead measure, Knot Count, Comparison Of Different	29-01-2026
6	**Software Testing and Quality Assurance:**	07
26	Software Testing and Quality Assurance: Concepts, Psychology of testing, Levels of testing	30-01-2026
27	Testing Process- test plan, test case design, Execution, Black Box testing , Boundary value analysis	05-02-2026
28	Pair wise testing- state based testing, White-Box testing criteria and test case generation and tool support	06-02-2026
29	Quality Assurance :Quality Control, Assurance, Cost,	19-02-2026
30	Reviews, Software Quality Assurance	20-02-2026
31	Software Quality Assurance, Approaches to SQA, Reliability,	26-02-2026
32	Quality Standards- ISO9000 And 9001	27-02-2026
7	**CASE Tools and Advance Practices of System Dependability and Security:**	05
33	Computer Aided Software Engineering Tools, SCRUM Developments,	28-02-2026
34	Dependable System	06-03-2026
35	Reliability Engineering,	12-03-2026
36	Safety Engineering,	13-03-2026
37	Security Engineering, Resilience Engineering	14-03-2026
8	**Advance Software Engineering:**	05
38	Software Reuse, Component Based Software Engineering,	19-03-2026

39	Distributed Software Engineering,	21-03-2026
40	Service-Oriented Software Engineering,	21-03-2026
41	Real-Time Software Engineering, Systems Engineering,	27-03-2026
42	Systems of System.	27-03-2026